

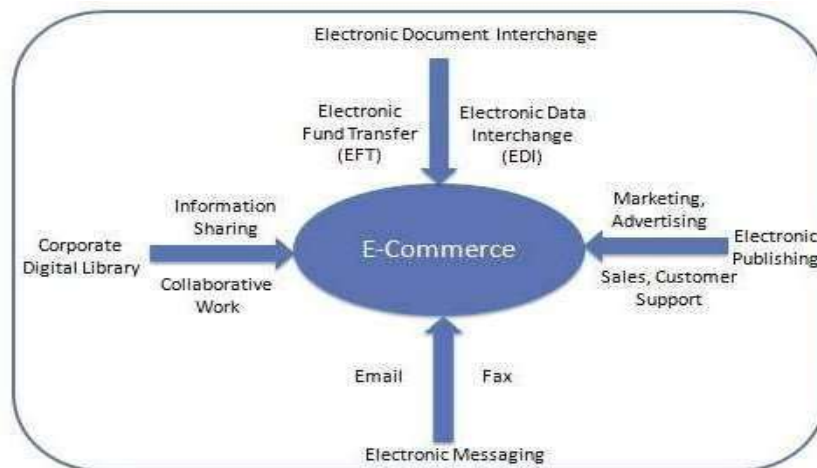
Unit 1: Introduction to Electronic-Commerce (E-commerce)

➞ 1.1 E-Commerce (EC) overview

- **Definition:**

“Electronic commerce or e-Commerce, consists of the buying and selling of products or services over electronic systems such as the Internet and other computer networks.”

- E-Commerce or Electronics Commerce is a methodology of modern business which addresses the need of business organizations, vendors and customers to reduce cost and improve the quality of goods and services while increasing the speed of delivery.
- E-commerce refers to paperless exchange of business information using following ways:
 - Electronic Data Exchange (EDI)
 - Electronic Mail (e-mail)
 - Electronic Bulletin Boards
 - Electronic Fund Transfer (EFT)
 - Other Network-based technologies



➡ 1.2 Features/Aims of E-commerce

E-Commerce provides following features:

- **Non-Cash Payment** – E-Commerce enables use of credit cards, debit cards, smart cards, electronic fund transfer via bank's website and other modes of electronics payment.
- **24x7 Service availability** – E-commerce automates business of enterprises and services provided by them to customers are available anytime, anywhere. Here 24x7 refers to 24 hours of each seven days of a week.
- **Advertising / Marketing** – E-commerce increases the reach of advertising of products and services of businesses. It helps in better marketing management of products / services.
- **Improved Sales** – Using E-Commerce, orders for the products can be generated anytime, anywhere without any human intervention. By this way, dependencies to buy a product reduce at large and sales increases.
- **Support** – E-Commerce provides various ways to provide pre sales and post sales assistance to provide better services to customers.
- **Inventory Management** – Using E-Commerce, inventory management of products becomes automated. Reports get generated instantly when required. Product inventory management becomes very efficient and easy to maintain.
- **Communication improvement** – E-Commerce provides ways for faster, efficient, reliable communication with customers and partners.

❖ Types of e-commerce

There are four category of E-Commerce:

- 1) **Business-to-Business [B2B]**
- 2) **Business-to-Consumer [B2C]**
- 3) **Consumer-to-Business [C2B]**
- 4) **Consumer-to-Consumer [C2C]**

❖ **Business-to-Business [B2B]**

- Business-to-Business [B2B] is an e-commerce in which one business deals with another business.
- This deal can be of products, goods or any services.
- For example: Google company tie-up with LG company to make a product as mobile.

❖ **Business-to-Consumer [B2C]**

- In this type of e-commerce a business can sale its products and services to the consumer/customer.
- For example: flipkart.com, amazon.com, www.icicibank.com.

❖ **Consumer-to-Business [C2B]**

- In this type of e-commerce a customer can work or provide services to business.
- For example: priceline.com

❖ **Consumer-to-Consumer [C2C]**

- This is an e-commerce where a consumer can sale a product to another consumer.
- For example: olx.com,quicker.com

➤ Advantages and Limitations of E-Commerce

- E-commerce benefits to E-Commerce advantages can be broadly classified in three major categories:
 - Advantages to Organizations
 - Advantages to Consumers
 - Advantages to Society

❖ Advantages to Organizations

- ✓ **Global Reach:** Using E-Commerce, organization can expand their market to national and international markets with minimum capital investment. An organization can easily locate more customers, best suppliers and suitable business partners across the globe.
- ✓ **Cost Reduction:** E-Commerce helps organization to reduce the cost to create process, distribute, retrieve and manage the paper based information by digitizing the information.
- ✓ **Better Customer Service:** E-commerce helps organization to provide better customer services also make high interaction with more customers.
- ✓ **Supply chain improvement:** E-Commerce improves the supply chain management by increased the productivity of the organization. It supports "pull" type supply management. In "pull" type supply management, a business process starts when a request comes from a customer and it uses just-in-time manufacturing way.
- ✓ **24*7 hour business:** The business is always open that is 24*7*365 days with no over time or extra time.
- ✓ **Rapid time to market:** EC provide quick implementation of an idea of the business because of consumer communication.
- ✓ **Efficient inventory management:** EC enables reduce administrative cost and reduce inventory management cost.

❖ Advantages to Customers

- ✓ **24x7 support:** Customer can do transactions for the product or enquiry about any product/services provided by a company anytime, anywhere from any location. Here 24x7 refers to 24 hours of each seven days of a week.
- ✓ **More products and services:** E-Commerce application provides user more options and quicker delivery of products.
- ✓ **Cheaper products and services:** E-Commerce application provides user more options to compare and select the cheaper and better option.
- ✓ **Review of product available:** A customer can put review comments about a product and can see what others are buying or see the review comments of other customers before making a final buy.
- ✓ **Readily available information:** A customer can see the relevant detailed information within seconds rather than waiting for days or weeks.
- ✓ **NO Sale Tax:** In many countries/cities online purchase does not have to pay sales tax.

❖ Advantages to Society

- ✓ **Telecommuting:** Customers need not to travel to shop a product thus less traffic on road and low air pollution.
- ✓ **Hope for the poor:** E-Commerce helps reducing cost of products so people of rural area can also afford the products.
- ✓ **Higher standard of living:** The living of people becomes standard by shopping online from the home.
- ✓ **Availability of public service:** E-Commerce helps government to deliver public services like health care, education, social services at reduced cost and in improved way.

➤ Disadvantages / Limitations of E-Commerce

E-Commerce disadvantages can be broadly classified in two major categories:

- Technical disadvantages
- Non-Technical disadvantages

Technical Disadvantages

- ✓ There can be lack of system security, reliability or standards owing to poor implementation of e-Commerce.
- ✓ Software development industry is still evolving and keeps changing rapidly.
- ✓ In many countries, network bandwidth might cause an issue as there is insufficient telecommunication bandwidth available.

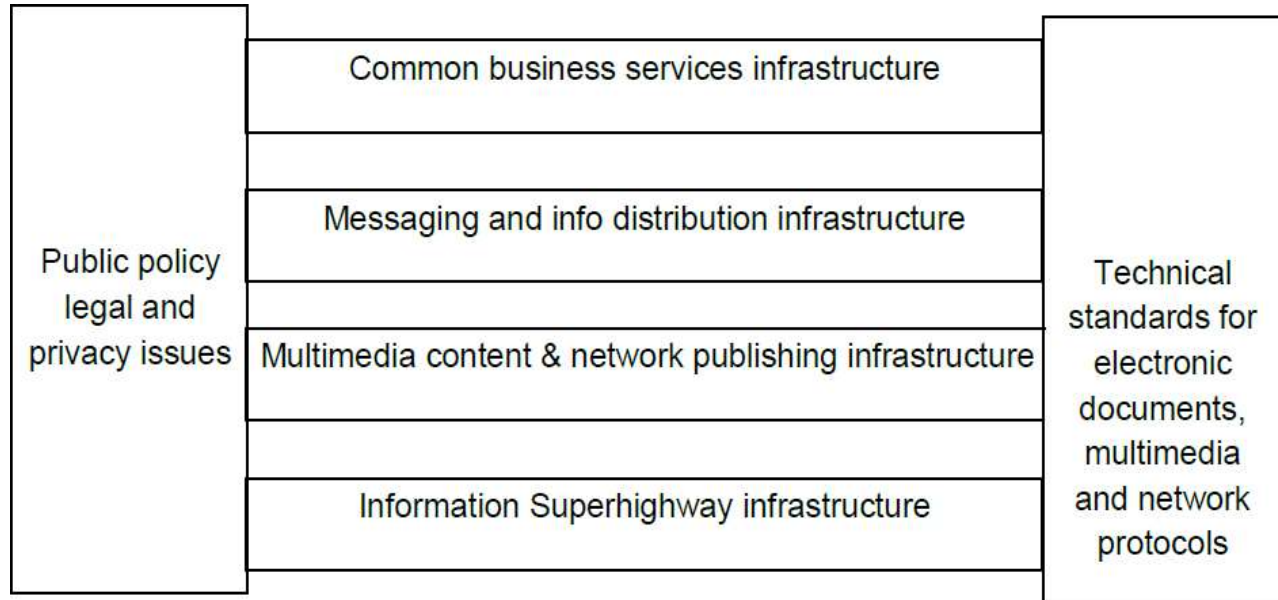
Non-Technical Disadvantages

- ✓ **Initial cost:** The cost of creating / building E-Commerce application in-house may be very high. There could be delay in launching the E-Commerce application due to mistakes, lack of experience.
- ✓ **User resistance:** User may not trust the site being unknown faceless seller. Such mistrust makes it difficult to make user switch from physical stores to online/virtual stores.
- ✓ **Product Quality:** Lack of touch or feel of products during online shopping.
- ✓ **Internet access** is still not cheaper and is inconvenient to use for many potential customers like one living in remote villages.

➤ 1.3 Framework of E-Commerce

- The E-commerce application will be built on the existing technology infrastructure like- computers, communication networks and communication software forming the information highway (i-way).

- There are two very important pillars to supports the applications of e-commerce in framework. These two pillars supporting all e-commerce applications and infrastructure:



1. Public Policy legal and privacy issues:

- This pillar has all the policy related to e-commerce for the business and also for the consumer.

2. Technical standard for electronic documents, multimedia and network protocols:

- This pillar provides the format for the information display can be use multimedia and also give the communication protocols for the network.
- The framework of e-commerce contains four types of Infrastructures:

1. Common Business services Infrastructure

- It provide the facility for buying and selling process.
- It includes security, authentication, e-payment etc.

2. Messaging and information distribution infrastructure

- It provide the facility for sending and receiving information.

3. Multimedia content and Network Publishing infrastructure

- It provide the facility to create information using Multimedia components like Audio, Video and Images.

4. Information superhighway infrastructure

- All the information in EC must travel using high-way.
- This information highway include telecom, cable TV, wireless, Internet etc.
- Any successful E-commerce Applications require this I-way infrastructure.
- The information must travel from this highway if it is business or it is for consumer.
- The information and Multimedia provide which type of information is passed. For example:

Digital games = Music + Video + Software

E-books = Text + Graphics + Data + Music + Photograph

- In the I-way the transportation of the product can be done using some types of software and network protocols.
- In a traditional transportation business, diesel engines or gasoline for powered motors to move the vehicle (truck) along the roadway.
- On the I-way, messaging software fulfills this role, in any number of forms like – e-mail, EDI or point-to-point file transfer.

➤ 1.4 Applications of E-Commerce

➤ Electronic Commerce Consumer Applications

- The wide range of applications for the consumer marketplace can be broadly classified into following:
 - ✓ **Entertainment:** Movies on demand, video cataloguing, interactive ads, multi user games and on line community.
 - ✓ **Financial services and information:** Home banking, financial services and financial news.
 - ✓ **Essential services:** Home shopping, electronic catalogs, telemedicine, remote diagnostics.

➤ 1.5 Electronic Commerce Organization Applications:

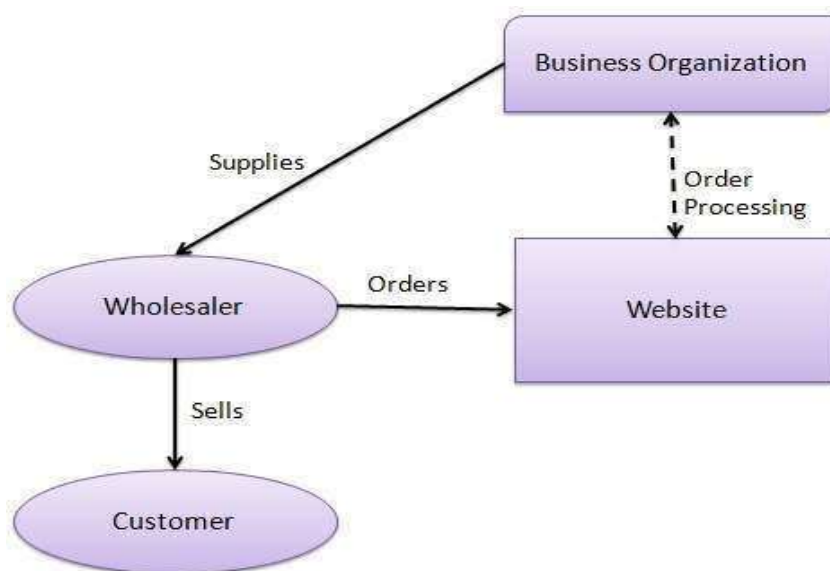
- The wide range of applications for the organization marketplace can be broadly classified into following
 - ✓ **Meeting customer needs and wants:** Analyzing customers' online behavior, trends and patterns and building a comprehensive customer database can lead to clearer understanding of how to attract new customers and retain existing ones.
 - ✓ **Delivering Products to customers:** In addition to developing e-commerce applications, packaging and distribution must be considered.
 - ✓ **Changing Business Environment:** Customers and businesses seek the flexibility to change trading partners, platforms, carriers and networks at will. Companies are looking outside and within when shaping business strategies.
 - ✓ **Marketing and Electronic Commerce:** E-com is forcing companies to rethink the existing ways of doing are: Target marketing (isolating and focusing on a segment of population). Relationship marketing (building and sustaining a long-term relationship with existing and potential customers). **"Virtual-Reality Inventory – Stock nothing, sell everything."**

♣ E-Commerce - Business Models

- E-Commerce or Electronics Commerce business models can generally categorized in following categories.
 - **Business - to - Business (B2B)**
 - **Business - to - Consumer (B2C)**
 - **Consumer - to - Consumer (C2C)**
 - **Consumer - to - Business (C2B)**

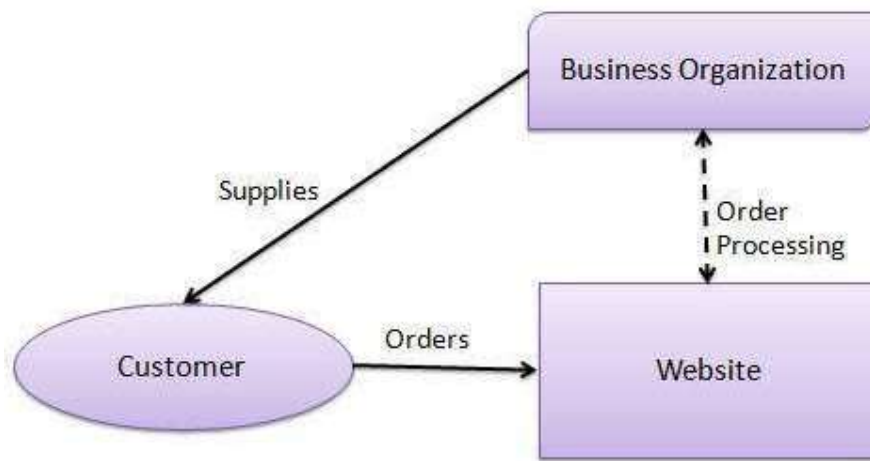
❖ **Business - to - Business (B2B)**

- ✓ A B2B model focuses on providing products from **one business to another**.
- ✓ Website following B2B business model sells its product to an intermediate buyer who then sells the product to the final customer.
- ✓ While many businesses like service providers are followed B2B model, you'll find software companies, office furniture and supply companies, document hosting companies etc..



❖ Business - to - Consumer (B2C)

- ✓ Website following B2C business model sells its product directly to a customer.
- ✓ A customer can view products shown on the website of business organization. The customer can choose a product and order the same.
- ✓ Website will send a notification to the business organization via email and organization will dispatch the product/goods to the customer.

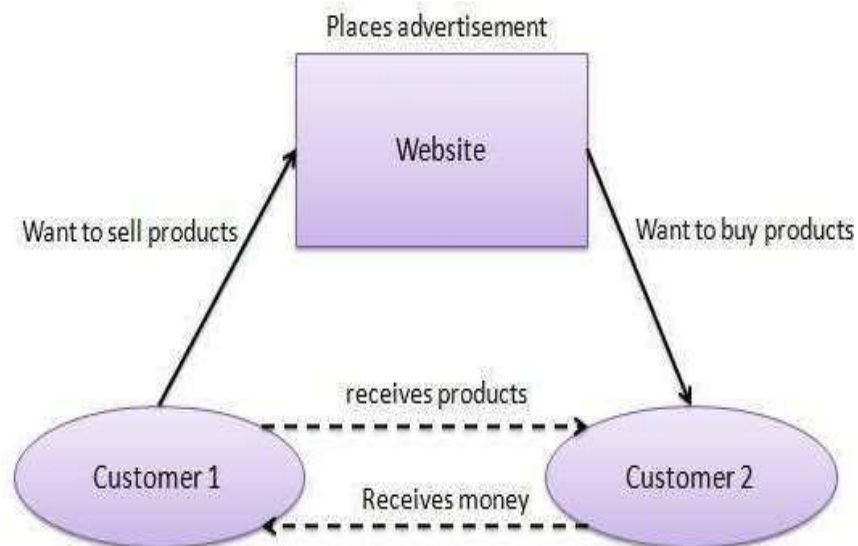


- ✓ Examples of B2C businesses are everywhere.
- ✓ Exclusively online retailers include flipkart.com, amazon.com

❖ Consumer - to - Consumer (C2C)

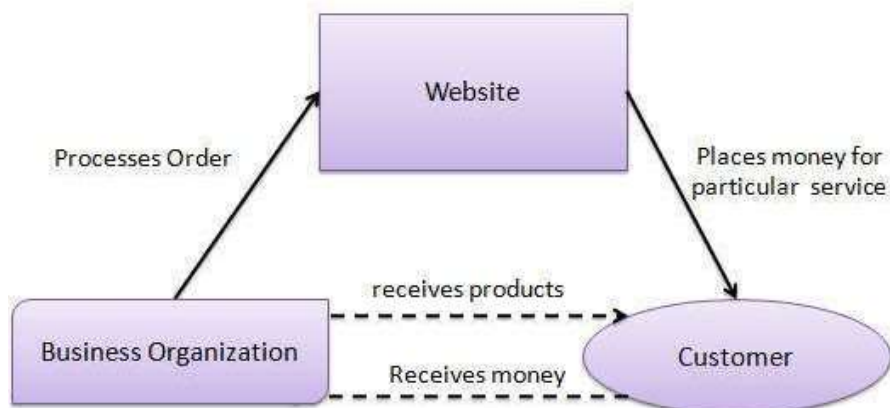
- ✓ Website following C2C business model helps consumer to sell their assets like residential property, cars, motorcycles etc. or rent a room by publishing their information on the website.
- ✓ Website may or may not charge the consumer for its services.
- ✓ Another consumer may opt to buy the product of the first customer by viewing the post/advertisement on the website.
- ✓ Created by the rise of the ecommerce sector and growing consumer confidence in online business, these ecommerce websites allow customers to trade, buy, and sell

Items in exchange for a small commission paid to the site. Opening a C2C site takes careful planning.



❖ Consumer - to - Business (C2B)

- In this model, a consumer approaches website showing multiple business organizations for a particular service. This online commerce business is when the consumer sells goods or services to businesses, and is roughly equivalent to a sole proprietorship serving a larger business. Consumer places an estimate of amount he/she wants to spend for a particular service. For example, comparison of interest rates of personal loan/ car loan provided by various banks via website. Eg. Google AdSense are example of C2B



♣ 1.6 M-Commerce

- Mobile commerce, also called M- Commerce, refers to any transaction completed with a mobile device – be it a smartphone, tablet, or even a wearable device.
- M-commerce (mobile commerce) is the buying and selling of goods and services through wireless handheld devices such as smartphones and tablets.
- It allows consumers to buy goods or services online – but from anywhere and at any time.

Examples of M-commerce innovations include:

- Online Tickets Booking,
 - Mobile Banking,
 - Money transfers, contactless payments, and in-app payments,
 - Digital content purchases,
 - Location-based services,
 - Mobile marketing, including coupons and loyalty cards.
- All of these m-commerce features which today are part of our customer experience wouldn't be possible if not for the rise of mobile commerce.

Types of mobile commerce applications

M-Commerce applications cover a broad range of transaction options. However, it's possible to divide them into one of the following types:

- **Mobile shopping** – this type of mCommerce implementation is most similar to eCommerce, only accessible through a mobile device. We're talking about dedicated apps, optimized websites, or even social media platforms like Facebook or Instagram that allow in-app purchases or linking to online stores.

- **Mobile banking** – mobile banking is quite similar to online banking. However, you will find that some transactions might be restricted or limited on mobile devices. Mobile banking is a type of implementation that usually involves a dedicated application – though some financial services companies are now experimenting with Chabot's or messaging apps to deliver customer service.
- **Mobile payments** – this type of mobile commerce focuses on the progressive innovation in payment options realized with mobile devices. Today consumers can take advantage of many diverse mobile payment options that go way beyond mobile wallets.